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RESEARCH ARTICLE

Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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Abstract

The ICH E17 guideline describes regional pooling in multi-regional clinical trials (MRCTs) in terms of the similarity of effect modifier distributions but names no measure, and to our knowledge no distance between regional effect modifier distributions has been evaluated for it. We quantify how two regions' distributions of each candidate effect modifier differ by the Wasserstein-1 (W_1) distance in original units. If the conditional average treatment effect is common to all regions, depends only on that modifier and is Lipschitz with clinical slope L_{clinical}, regional risk or mean differences differ by at most $\Delta_{\max} = L_{\text{clinical}} W_1$; when L_{clinical} is unknown, the smallest slope L^* at which the distance could produce a clinical margin is compared with a sponsor's bound. In a selection simulation (matches supplied), the standardized mean difference stayed at chance up to 2,000 patients per region in some configurations; W_1 and the Kolmogorov–Smirnov (KS) statistic reached the target (AUC 0.9) in every configuration, W_1 needing fewer patients in 11 of 13 cells (three by construction), overlapping in one, and more in the location-shift control on contaminated tails. For true distances of about 0.4 to 1.2 SD (normal reference; equal n), bootstrap coverage from 100 patients per region was 0.92–0.95, below 0.95 beyond Monte Carlo error in 5 of ten cells (0.947–0.951 at the application's operating point). In a GUSTO-I-based example, partner rankings on age and systolic blood pressure differed; a null floor marked distances not distinguishable from no difference. The framework quantifies differences; the pooling judgment remains clinical.

KEYWORDS
multi-regional clinical trial; ICH E17; effect modifier; Wasserstein distance; regional pooling; distributional similarity

1 | INTRODUCTION

Multi-regional clinical trials (MRCTs), conducted across multiple countries or regulatory regions under a single protocol, have become a common design in global pharmaceutical development, and methods for assessing the consistency of treatment effects across their regions have developed with them.[1,2] The International Council for Harmonisation (ICH) E17 guideline, adopted in 2017, established principles for planning and designing MRCTs to evaluate whether treatment effects can be regarded as generalizable across the target population.[3]

Regulatory authorities in Japan and China examine the consistency of treatment effects in regional subpopulations,[4,5] yet individual regional sample sizes in global MRCTs are often insufficient for that purpose. Regional pooling, in which regions with similar patient characteristics are grouped for analysis, is a key strategy for enhancing the assessment of regional consistency, and the ICH E17 guideline describes pooling decisions in terms of the similarity of effect modifier distributions.[3]

An effect modifier is a baseline patient characteristic, such as age, disease severity or a genetic marker, for which the treatment benefit differs across subgroups. Where it exists, regions with different patient compositions may observe different average

Abbreviations: AMI, acute myocardial infarction; AUC, area under the receiver operating characteristic curve; CATE, conditional average treatment effect; CDF, cumulative distribution function; CI, confidence interval; CRP, C-reactive protein; CV, coefficient of variation; EDF, empirical distribution function; GUSTO, Global Utilization of Streptokinase and t-PA for Occluded coronary arteries; ICH, International Council for Harmonisation; IPD, individual patient data; KL, Kullback–Leibler; KS, Kolmogorov–Smirnov; MRCT, multi-regional clinical trial; MRI, magnetic resonance imaging; NMPA, National Medical Products Administration; RMSE, root mean squared error; RWE, real-world evidence; SBP, systolic blood pressure; SD, standard deviation; SMD, standardized mean difference.

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treatment effects even if the drug works identically at the individual level: a region with predominantly younger patients would show larger benefits than one with predominantly older patients because the patient mix differs, not the drug. The similarity of the full effect modifier distribution, not only its average level, therefore matters for regional pooling. We focus on continuous effect modifiers such as age and disease severity scores; categorical effect modifiers raise distinct considerations and are outside the present scope.

Despite the regulatory importance of effect modifier distributional similarity, current practice lacks a standardized, clinically calibrated methodology. The ICH E17 guideline provides no specific metric, threshold, or statistical procedure for determining when distributions are "similar enough." Song et al., on the NMPA's implementation of ICH E17, identify the pooling strategy as an area where ambiguity remains and describe, when several effect modifiers are considered, clustering populations on distances defined from the correlation between effect modifiers and endpoints.[5] The pooling-strategy chapter of a reference volume on multi-regional trials under ICH E17 introduces, as one of many possible approaches, cluster analysis of regions on region-level summaries of the candidate effect modifiers: each region is given a representative value for each candidate, the values are typically standardized, and distances such as the Euclidean distance are weighted for relevance, for instance through a Lasso regression on the outcome; the chapter notes that the selection of the distance measure is one of the key decisions, and does not evaluate the approach.[6] Two gaps remain. First, to our knowledge no distance between regional effect modifier distributions has been proposed and evaluated for this purpose: a single summary per modifier leaves two regions that agree on it but differ in spread, shape, or tail behaviour at distance zero—for a binary characteristic, as in the chapter's illustration, one proportion describes the distribution completely, so the limitation concerns continuous modifiers—and any finite set of summaries requires deciding in advance which distributional features are relevant. Second, the chapter calls for a judgment of whether differences in the impact of effect modifiers across pooled regions are "clinically meaningful or not" but supplies no metric to operationalize that judgment.

The ingredients are not new in themselves. The inequality on which the clinical calibration below rests is the Kantorovich–Rubinstein duality, and the same bound on a difference of expectations under a Lipschitz function is used in the optimal-transport analysis of domain adaptation,[7] and a Wasserstein-1 bound of the same family holds under a reproducing-kernel condition;[8] the similarity between a trial sample and a target population has been summarized by propensity-score-based metrics[9] and by a generalizability index,[10] within the wider literature on extending trial inferences to a target population;[11] and the similarity of two distributions within a trimmed Mallows-distance neighbourhood has been framed as an equivalence-testing problem.[12] None of these converts the difference between two regions' distributions of an individual effect modifier into a bound on the regional effect difference whose constant is a clinical slope, or evaluates such a distance for regional pooling. The contribution of this paper is the formulation for regional pooling in MRCTs, the reverse calibration L^* when the slope is unknown, and the evaluation of the estimator's identification and estimation properties.

In the PREVENT trial of secukinumab in non-radiographic axial spondyloarthritis, a workshop report by Matsushima et al. describes a post hoc pooling in which countries with a similar distribution of effect modifiers were clustered into five pooled regions, with treatment differences that varied slightly with the distribution of CRP and MRI positivity and sex across them; the report attributes the inconsistent result in the Japanese subgroup to imbalanced treatment assignment and a small sample size rather than to effect modifiers.[4] Forming pooled regions from effect modifier distributions is the step that a distance between regional distributions would quantify; that case involved binary characteristics, whereas continuous effect modifiers can differ in shape and spread as well as in level.

General-purpose comparison tools such as the standardized mean difference are routinely applied to baseline covariates,[13] but the standardized mean difference captures only differences in location: regions can match on means yet differ in variance or distributional shape. When effect modification is non-linear, these features directly influence regional average treatment effects.

This paper addresses these gaps by proposing a quantitative framework for assessing effect modifier distributional similarity across regions, based on the Wasserstein-1 (W_1) distance between two cumulative distribution functions, retained in the original measurement units of the effect modifier. We estimate W_1 with bootstrap confidence intervals and link it to clinical interpretation through a calibration framework.

The remainder of this paper is organized as follows. Section 2 presents the framework, Section 3 two simulation studies of identification and estimation, Section 4 an illustration with publicly available individual patient data from the GUSTO-I trial, and Section 5 implications, limitations, and future directions.

2 | METHODS

Let the conditional average treatment effect (CATE) be denoted $\tau(x) = E[Y(1) - Y(0) \mid X = x]$, where X is the effect modifier. When this function is non-constant, the average treatment effect observed in region r depends on the distribution F_r of the effect modifier in that region:

$$\bar{\tau}_r = \int \tau(x)\, dF_r(x). \tag{1}$$

Assessing pooling suitability therefore requires a measure of distributional difference between regions. Equation (1), and everything built on it, presupposes an effect scale on which the regional effect is the average of the conditional effects, that is, a collapsible contrast such as a risk difference, a mean difference or a difference in restricted mean survival time; marginal odds and hazard ratios are not averages of their conditional counterparts and fall outside Proposition 1 unless the regional effect is first defined by standardization on a collapsible scale.

2.1 | Existing Approaches and Their Limitations

The most widely used measure for comparing covariate distributions across groups is the standardized mean difference (SMD), defined as

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_2}{S_{\text{pooled}}}, \tag{2}$$

where $\bar{X}_r$ denotes the sample mean in region r and S_{pooled} is the pooled standard deviation.[13] By construction, SMD captures only differences in location. Two distributions with identical means but markedly different variances—for example, $N(50, 5^2)$ and $N(50, 15^2)$—yield SMD = 0 despite a threefold difference in spread.

Distributional distance measures address this limitation by comparing distributions beyond summary statistics. Empirical distribution function (EDF) tests such as the Kolmogorov–Smirnov (KS) statistic, $D_{\text{KS}} = \sup_x |F_1(x) - F_2(x)|$, and related measures (Cramér–von Mises, Anderson–Darling) quantify differences in the full shape of distributions by comparing CDFs directly. Information-theoretic divergences such as Kullback–Leibler (KL) divergence,

$$D_{\text{KL}}(P\|Q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx, \tag{3}$$

take a density-based approach, measuring how one probability distribution differs from another.

Each of these approaches has limitations for comparing effect modifier distributions across regions. SMD captures only location and is blind to variance and shape. EDF-based statistics such as the KS statistic capture full distributional differences but lack a clinically interpretable scale, and the bound on treatment effect heterogeneity they admit is governed by the total variation of the CATE, a constant that is clinically stated only for a monotone CATE (Section 2.2). KL divergence is asymmetric, can diverge to infinity when empirical distributions do not fully overlap, and requires density estimation that is impractical with small regional samples. None of these measures offers a bound on regional treatment effect differences whose constant is a per-unit clinical slope, the kind of quantity that treatment-by-covariate interaction analyses estimate. These gaps motivate a measure that (i) captures distributional features beyond location, (ii) provides a clinically interpretable scale, and (iii) offers such a bound; the Wasserstein-1 distance and its clinical calibration, developed next, satisfy all three.

2.2 | Per-Effect-Modifier Wasserstein-1 Distance

Among distributional distances that could capture the full distributional differences relevant to regional pooling, we adopt the Wasserstein-1 distance (W_1) for three reasons: it responds to differences beyond location; the heterogeneity bound it admits (Proposition 1 below) has a per-unit clinical slope L_{clinical} as its constant; and that bound is valid for every CATE in the class, sharp when the slope is the only restriction placed on the CATE, and tighter, as a function of W_1 and the mean difference, when the CATE is also monotone; for any particular CATE it is therefore a conservative upper bound. The W_1 distance (sometimes called the Earth Mover's Distance) admits three equivalent representations on the real line, each highlighting a different facet of its interpretation.[14,15]

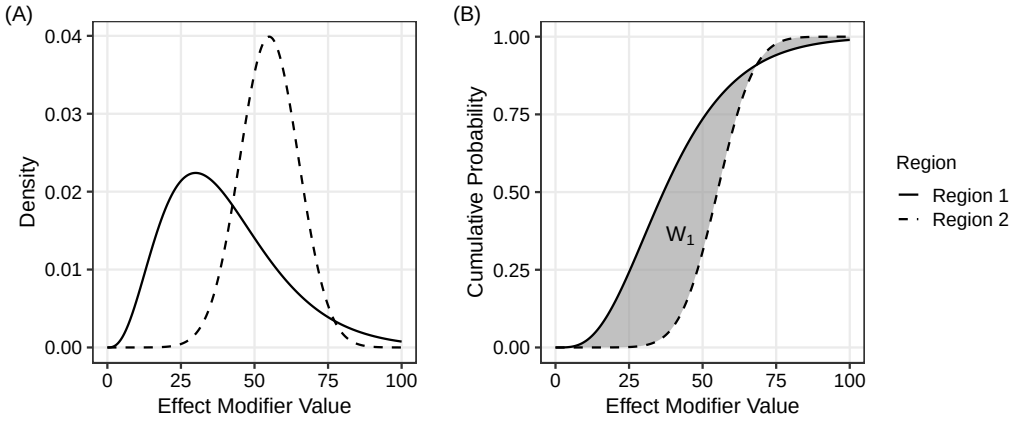


FIGURE 1 The Wasserstein-1 distance as the area between cumulative distribution functions, illustrated with a right-skewed Region 1 (gamma distribution) and a symmetric Region 2 (normal distribution). Panel (A): probability densities. Panel (B): cumulative distribution functions; the shaded area equals $W_1(F_1, F_2)$.

Let F_1 and F_2 denote the cumulative distribution functions (CDFs) of an effect modifier in two regions. Then

$$W_1(F_1, F_2) = \int_{-\infty}^{\infty} |F_1(x) - F_2(x)|\, dx = \int_0^1 |F_1^{-1}(u) - F_2^{-1}(u)|\, du = \inf_{\pi \in \Pi(F_1, F_2)} \int |x - y|\, d\pi(x, y), \tag{4}$$

where F_r^{-1} is the quantile function and $\Pi(F_1, F_2)$ is the set of probability couplings with marginals F_1 and F_2. The CDF form expresses W_1 as the area between the two CDFs (Figure 1); the quantile form as the integrated absolute difference of quantile functions; and the Kantorovich form as the minimum expected absolute displacement to transport one distribution onto the other. The three forms are mathematically equivalent.[15] Unlike the standardized mean difference, W_1 responds to differences in variance, skewness, and other distributional features.[16]

W_1 carries the units of the underlying effect modifier in each of the three equivalent forms of equation (4): the CDF form integrates a dimensionless difference of probabilities over x, the quantile form differences of quantiles, and the coupling form distances $|x - y|$. $W_1(F_1, F_2)$ is therefore expressed on the original scale of the effect modifier, which supports direct clinical interpretation: a regional $W_1 = 5$ mmHg in systolic blood pressure means that optimally transporting one distribution onto the other moves probability mass an average distance of 5 mmHg. The same logic precludes a meaningful direct comparison of W_1 values across effect modifiers measured on different scales, so we restrict attention to per-effect-modifier assessment (Section 5 elaborates on this scope choice).

W_1 is a proper metric on the space of probability distributions with finite first moment, satisfying non-negativity, symmetry, identity of indiscernibles, and the triangle inequality.[14] It is also affine-equivariant in the variable units, satisfying $W_1(F_{aX+b}, F_{aY+b}) = |a| \cdot W_1(F_X, F_Y)$ for any $a \neq 0$. Beyond these basic properties, W_1 admits a Kantorovich–Rubinstein dual representation[14]

$$W_1(F_1, F_2) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left| \int f\, dF_1 - \int f\, dF_2 \right|, \tag{5}$$

which expresses W_1 as the largest possible difference in expected values of a 1-Lipschitz function f between the two distributions (the notation $\|f\|_{\text{Lip}} \leq 1$ means $|f(x) - f(y)| \leq |x - y|$ for all x, y). If the conditional average treatment effect function $\tau(x)$ is Lipschitz continuous with constant L_{clinical}, so that $|\tau(x) - \tau(y)| \leq L_{\text{clinical}}|x - y|$ for all x, y, then τ/L_{clinical} is 1-Lipschitz and applying equation (5) yields the central result of this section:

Proposition 1. (Connection to heterogeneity). *Suppose the CATE function τ is the same in every region and depends on the effect modifier X alone. If τ is Lipschitz continuous with constant $L_{clinical}$, then for any two regions with effect modifier distributions F_1 and F_2,*

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq L_{clinical} \cdot W_1(F_1, F_2). \tag{6}$$

Proof. Apply equation (5) with $f = \tau/L_{\text{clinical}}$, which is 1-Lipschitz by assumption. This yields $\left|\int (\tau/L_{\text{clinical}})\, d(F_1 - F_2)\right| \leq W_1(F_1, F_2)$. Combining with equation (1) for $\bar{\tau}_r$ and rearranging yields equation (6). □

Equation (6) is not the only bound of its form. Integration by parts gives $\bar{\tau}_1 - \bar{\tau}_2 = -\int (F_1(x) - F_2(x))\, d\tau(x)$, so for a CATE of bounded variation on a bounded support, $|\bar{\tau}_1 - \bar{\tau}_2| \leq \text{TV}(\tau) \cdot D_{\text{KS}}$ as well, and $W_1 \leq W_2$ makes $L_{\text{clinical}} \cdot W_2$ a valid but looser bound.[14] Equation (6) is sharp when the Lipschitz constant is the only restriction placed on the CATE: by the Kantorovich–Rubinstein duality of equation (5), $L_{\text{clinical}} W_1$ is the supremum of $|\bar{\tau}_1 - \bar{\tau}_2|$ over all common L_{clinical}-Lipschitz CATEs, attained by $\tau^*(x) = L_{\text{clinical}} \int_{x_0}^{x} \text{sgn}(F_2(t) - F_1(t))\, dt$, whose slope changes sign at every crossing of the two CDFs; the Kolmogorov–Smirnov bound is likewise sharp within its own class (Appendix A.3). Sharpness over a class is a property of the class, and shape information available to the sponsor tightens the bound. If τ is also monotone, with $0 \leq \tau' \leq L_{\text{clinical}}$, then $\bar{\tau}_1 - \bar{\tau}_2 = \int (F_2 - F_1)\, \tau'\, dx$ lies between $-L_{\text{clinical}} A_-$ and $L_{\text{clinical}} A_+$, where $A_\pm = \int (F_2 - F_1)_\pm\, dx$ satisfy $A_+ + A_- = W_1$ and $A_+ - A_- = \mu_1 - \mu_2$, so that

$$\sup_{\tau \text{ monotone, } L_{\text{clinical}}\text{-Lipschitz}} |\bar{\tau}_1 - \bar{\tau}_2| = L_{\text{clinical}} \max(A_+, A_-) = \tfrac{1}{2} L_{\text{clinical}} \left(W_1 + |\mu_1 - \mu_2|\right), \tag{7}$$

which equals $L_{\text{clinical}} W_1$ when the CDFs do not cross and $L_{\text{clinical}} W_1/2$ when the means coincide. A bound on the magnitude of τ itself tightens it further, and over the recorded supports of Section 4, at the illustrative bounds L_{UB} of Section 4.4, the attaining τ^* would require effects beyond the baseline risks (Appendix A.3). $\Delta_{\max} = L_{\text{clinical}} W_1$ is therefore a conservative upper bound by construction; Section 4.4 reports the factor by which equation (7) would shrink it in the application. Sharpness, which the Kolmogorov–Smirnov bound also has within its own class, does not distinguish the two bounds; the constant does, L_{clinical} being a change in treatment effect per unit of the effect modifier, the kind of quantity that interaction analyses estimate, whereas $\text{TV}(\tau)$ is clinically stated only for a monotone CATE. Neither the W_1 bound nor the Kolmogorov–Smirnov bound dominates the other in general (Appendix A.3).

The right-hand side of equation (6) involves W_1 in its original measurement units multiplied by a clinical slope L_{clinical}, so no further normalization is required for a clinically interpretable bound and we adopt W_1 in its original units throughout; normalizing by a within-effect-modifier scale measure such as the pooled interquartile range is redundant once the bound is used for calibration (Section 2.4). The Lipschitz assumption underlying Proposition 1 is appropriate when the CATE function varies smoothly with the effect modifier; for threshold or step-function CATEs, L_{clinical} can be arbitrarily large and the bound becomes conservative (Section 5).

2.3 | Estimation

Given samples $\{X_{1,i}\}_{i=1}^{n_1}$ from region 1 and $\{X_{2,j}\}_{j=1}^{n_2}$ from region 2, the Wasserstein-1 distance is estimated using empirical distribution functions. Since $\hat{F}_1$ and $\hat{F}_2$ are piecewise constant (right-continuous step functions), the integral in equation (4) reduces to a finite sum over the intervals between consecutive distinct values of the pooled sample:

$$\widehat{W}_1 = \sum_{k=1}^{m-1} |\hat{F}_1(x_{(k)}) - \hat{F}_2(x_{(k)})| \cdot (x_{(k+1)} - x_{(k)}), \tag{8}$$

where $\hat{F}_r$ denotes the empirical CDF (right-continuous) of region r and $x_{(1)} < \cdots < x_{(m)}$ are the m distinct values of the pooled sample (ties are pervasive for coarsely recorded modifiers, such as systolic blood pressure in whole mmHg in Section 4). The asymptotic distribution of W_1 is non-standard: del Barrio et al.[17] showed that, when $\int \sqrt{F(1-F)}\, dx < \infty$, $\sqrt{n}\, W_1(\hat{F}_n, F) \xrightarrow{d} \int |B(F(x))|\, dx$, where B is a standard Brownian bridge. Its quantiles depend on the unknown F, so no universal critical values are available. In the two-sample case a Gaussian limit, and hence a Wald interval, exists under the measure-zero condition stated below; the same condition governs the consistency of the bootstrap, so the choice between them rests on finite-sample behaviour: the Wald interval requires an estimate of the limit's variance, which depends on $\text{sgn}(F_1 - F_2)$ and hence on where the CDFs cross, whereas the percentile bootstrap requires no variance estimate. We therefore employ the nonparametric percentile bootstrap with $B = 2{,}000$ replicates.[18] Each bootstrap replicate draws n_1 observations with replacement from region 1 and, independently, n_2 from region 2, and recomputes $\widehat{W}_1$ by equation (8); the 95% interval is formed by the 2.5th and 97.5th sample percentiles of the B replicates (R's default quantile definition, type 7). Under the conditions of Appendix A.2—integrable $\sqrt{F_r(1-F_r)}$, a sample ratio $n_1/(n_1+n_2)$ away from 0 and 1, and $\{x : F_1(x) = F_2(x) \in (0,1)\}$ of Lebesgue measure zero, as when the two CDFs cross only

at isolated points—the Hadamard derivative of the L_1 functional at $F_1 - F_2$ is linear and the ordinary nonparametric bootstrap is consistent by the functional delta method;[19] where the CDFs coincide on an interval, including $F_1 = F_2$, the derivative is non-linear and the naive bootstrap is not consistent,[20,19,21] and convergence occurs at rate $\sqrt{n_1 n_2/(n_1 + n_2)}$. The estimator (8) is computed in $O((n_1 + n_2)\log(n_1 + n_2))$ time, dominated by sorting the combined order statistics.

Because W_1 is a distance, $\widehat{W}_1$ is non-negative by construction and is typically positive even when the two regions are draws from the same distribution. More precisely, it is biased upward at every sample size, $E\,\widehat{W}_1 \geq W_1$ (Appendix A.1), and the percentile interval does not correct this bias, so near the boundary the whole interval is displaced upward (Section 3.2.2). For the patients actually enrolled, on the other hand, the bound holds exactly with the empirical distributions: the difference between the two enrolled samples' average treatment effects is at most $L_{\text{clinical}}\, W_1(\hat{F}_1, \hat{F}_2)$, whatever the sampling error of $\widehat{W}_1$ as an estimate of W_1. We refer to the sampling distribution of $\widehat{W}_1$ under $F_1 = F_2$ as the null floor. It is obtained by drawing two independent resamples, of the two regions' actual sizes, from one region's empirical distribution—the anchor's in Section 4.2—and recomputing $\widehat{W}_1$; because it depends on both sample sizes and on the source distribution, it is computed per region pair. An observed $\widehat{W}_1$ at or below its 95th percentile, the reference point used in Section 4.2, is one the estimator would produce with appreciable probability from identical distributions, so it is not evidence of a difference; how large the difference could be is read from the upper confidence limit of $\widehat{W}_1$, equivalently of $\Delta_{\max}$. Declaring a distance unresolved in this way is a one-sided test of $F_1 = F_2$ at level 0.05; it is used only to mark which estimates are read at their upper limit, and the comparisons of Section 4 are not adjusted for multiplicity, since no inference about pooling is drawn from the number of unresolved distances.

2.4 | Interpretation and Clinical Calibration

We refer to the Lipschitz constant of the CATE function $\tau(x)$ as the clinical slope L_{clinical}. By definition, L_{clinical} quantifies the maximum change in the treatment effect per unit change in the effect modifier, expressed on the outcome scale of the trial—for example, absolute 30-day mortality risk per year of age, or per mmHg of systolic blood pressure. Throughout this paper L_{clinical} is quoted on the proportion scale of the outcome, so that $L_{\text{clinical}} = 1 \times 10^{-2}$ per year denotes one percentage point of absolute 30-day mortality per year of age; clinical margins Δ_{clin} stated in percentage points are divided by 100 before entering any calculation. When prior individual patient data (IPD) are available, meta-regression on treatment-by-covariate interaction estimates the average interaction slope.[22,23] An average slope is, in absolute value, a lower bound on the Lipschitz constant rather than the constant itself: for non-linear τ, the local slope can exceed the fitted one. An L_{clinical} specified from such an estimate should therefore carry a sponsor-specified multiplicative safety factor, in the spirit of margin discounting in non-inferiority design, so that the premise of Proposition 1 remains defensible. More commonly at the MRCT planning stage, published subgroup reports document the direction of interaction rather than a per-unit slope, and a precise numerical L_{clinical} is unavailable. We address both situations below.

From Proposition 1, the maximum potential difference in regional average treatment effects attributable to effect modifier distributional differences is

$$\Delta_{\max} = L_{\text{clinical}} \cdot W_1(F_1, F_2), \tag{9}$$

expressed in the units of the trial outcome scale. The formula is invariant to within-effect-modifier scale normalization of W_1: any normalized form W_1/X multiplied by the same scale X cancels in equation (9), so adopting W_1 in its original units yields the same clinical-scale calibration as any rescaled version. When L_{clinical} is independently known, equation (9) provides a direct clinical-scale translation of the observed distributional difference. When L_{clinical} is unknown, the relation can be inverted to ask which L_{clinical} would be required for a given W_1 to produce a clinically meaningful $\Delta_{\max}$.

The estimator $\widehat{W}_1$ has a percentile bootstrap CI (Section 2.3), and inferential uncertainty propagates linearly to $\Delta_{\max}$ when L_{clinical} is treated as fixed: $[\Delta_{\max,L}, \Delta_{\max,U}] = L_{\text{clinical}} \cdot [W_{1,L}, W_{1,U}]$. When L_{clinical} itself is uncertain or unavailable, we solve equation (9) for L_{clinical} at a pre-specified clinical margin Δ_{clin} (the minimal clinically important treatment effect difference, the non-inferiority margin, or the targeted overall effect), yielding the required clinical slope

$$L^* = \frac{\Delta_{\text{clin}}}{W_1(F_1, F_2)}. \tag{10}$$

L^* is the smallest clinical slope under which the observed distributional difference could produce a clinically meaningful regional treatment effect difference, within the class of CATEs restricted only by their slope. Because L^* is a decreasing function of W_1, the bootstrap CI of $\widehat{W}_1$ gives the percentile interval $[\Delta_{\text{clin}}/W_{1,U},\ \Delta_{\text{clin}}/W_{1,L}]$ for L^*. If a plausible upper bound on L_{clinical} from prior evidence (denoted L_{UB}) lies below L^*, the distributional difference cannot produce such a difference, provided

$L_{\text{clinical}} \leq L_{\text{UB}}$; reading L_{UB} against the lower limit $\Delta_{\text{clin}}/W_{1,U}$ rather than the point value carries the estimation uncertainty of $\widehat{W}_1$ into this comparison.

Specifying L_{clinical} (or its upper bound L_{UB}) is a clinical judgment, informed but not determined by prior evidence, for which the sponsor carries responsibility—the same footing as a non-inferiority margin. Writing $R_a(x)$ for the risk under arm a, $\tau(x) = R_1(x) - R_0(x)$, so $|\tau'(x)| \leq |R_1'(x)| + |R_0'(x)|$: arm-level risk gradients constrain the modification slope only jointly, and a prognostic gradient alone guarantees no bound, prognostic and treatment-effect gradients being distinct quantities.[24,25] For the GUSTO-I case study presented in Section 4, the Fibrinolytic Therapy Trialists' meta-analysis[26] and the GUSTO-I prognostic model[27] inform illustrative bounds of $L_{\text{UB,age}} \approx 1 \times 10^{-2}$ per year and $L_{\text{UB,SBP}} \approx 2 \times 10^{-3}$ per mmHg on the 30-day mortality scale; these exceed the average prognostic age gradient implied by the 30-day mortality that Lee et al. report for GUSTO-I (at most about 6.5×10^{-3} per year; Appendix A.3),[27] though not the sum of two arm-level gradients of that size that the inequality above permits—a reference point rather than a guarantee. The age bound is also large relative to the overall treatment effect in GUSTO-I, a 30-day mortality of 6.3% under accelerated t-PA against 7.2% and 7.4% under the two streptokinase-only strategies,[28] which is one reason the bounds are illustrative. Conclusions drawn from the bounds are conditional on them; because L^* is reported for every region pair, any alternative bound can be compared against it directly.

The pooling assessment proceeds by computing per-effect-modifier W_1 with bootstrap CIs across candidate partner regions, ordering the candidates by the point estimate $\widehat{W}_1$ within each effect modifier, reading each distance against its null floor and its upper confidence limit (Section 2.3)—distances unresolved against the null floor are not ordered among themselves—and deriving $\Delta_{\max}$ when L_{clinical} is known or L^* at a pre-specified Δ_{clin} when L_{clinical} is unknown. The framework is deliberately estimation-centered rather than testing-based: W_1 has no universal cutoff that separates "similar" from "dissimilar," and what a given W_1 means clinically depends on the clinical slope L_{clinical}, which is external to the distributional comparison. Per-pair $\Delta_{\max}$ (or L^*) with bootstrap CIs, presented alongside the clinical context (the overall treatment effect size, the non-inferiority margin, the minimal clinically important difference), are quantitative inputs for sponsor and regulatory judgment, which is not driven by a significance test;[29] the one test, the null floor of Section 2.3, governs how an estimate is displayed and read, not whether regions are pooled.

3 | SIMULATION STUDIES

We conducted two complementary simulation studies. Study 1 evaluates identification: whether each candidate distance measure can detect, at some sample size, that a candidate region's effect modifier distribution differs from the anchor's, scored on a selection task in which candidates built from the anchor's population are ranked against candidates built from a different population and the number of true matches is supplied. It does not evaluate the ordering of regions that all differ from the anchor, which is what the application reports, and it does not rank regions by their effect difference, which depends on the unknown CATE: under a linear CATE only the mean difference matters, so a region differing from the anchor in scale alone has an effect difference of zero although $W_1 > 0$, whereas under the V-shaped CATE $\tau(x) = L_{\text{clinical}}|x - 50|$ the same scale difference produces the full $L_{\text{clinical}} W_1$ (in scenario S5 of Study 2, $L_{\text{clinical}}\sqrt{2/\pi} \cdot 5$) while the SMD is zero. The Lipschitz class contains both CATEs, Proposition 1 covers both, and the SMD covers only the first; Study 1 asks which measures are blind to differences of the second kind. Study 2 evaluates estimation performance: bias, variability, and coverage probability of bootstrap confidence intervals for $\widehat{W}_1$. Study 1 carries the comparison between measures, because a distance that cannot distinguish differing distributions cannot usefully be calibrated; Study 2 underwrites the calibration pathway that converts an estimated distance into a $\Delta_{\max}$ statement via equation (9).

Study 1's required sample sizes are discrimination statements, the per-region sample size at which a measure's area under the receiver operating characteristic curve (AUC) reaches at least 0.9, which for $\widehat{W}_1$ ranges from 42 to 690 across the cells of Sets 3 and 4; Study 2's estimation statements (Section 3.2.2) are not interchangeable with these.

3.1 | Study 1: Identification

3.1.1 | Design

Study 1 follows the ADEMP structure (aims, data-generating mechanisms, estimands, methods, performance measures) for simulation studies.[30] Four worlds are simulated, each built from a single distributional family and each containing regions

constructed either from the anchor's own population (true matches) or from a discordant population. Set 1 (Gaussian) places discordance in location and scale. Set 2 (log-normal) places it in location and in dispersion combined with shape. Set 3 (mixture) is the adversarial world for summary-based measures: every region, matched or discordant, is constructed with mean 50 and SD 10, so all discordance lies in distributional shape. Set 4 (extremes) places a 5% subgroup in each tail, 40 units from the centre in the anchor; discordant regions displace both subgroups (to 70 or 100 units), raise their prevalence to 12% (mean fixed), or displace only the upper subgroup to 100 units; two control regions instead shift their whole population by 2 and 3 units. In the three Set 4 cells in which a subgroup is displaced (symmetric and asymmetric severity) or its prevalence raised, the two full-distribution measures respond differently by construction: a subgroup of prevalence ε displaced by d changes the KS statistic by at most about ε whatever d is, while it changes W_1 by εd, and mass added 40 units from the centre moves W_1 by that distance per unit of added mass, so W_1's sample-size advantage in those cells follows from the mechanism, and the bulk-shift control, a location shift of the same population, is the cell that removes it. Ground truth is the construction label—which regions were built from the anchor's population—so the target is defined without reference to any of the compared measures and the comparison is not circular.

Three measures enter the comparison: W_1; the KS statistic, the most common comparison of full distributions; and SMD, the standard comparison of location. Under the registered protocol the simulation also evaluated three representative-value distances built from one to three regional moments, SMD computed on the log scale in the log-normal world (which recovers the shape contrast, AUC 1.00 at n = 2,000, but leaves the combined cell at chance, 0.51), a harm measure based on the largest W_1 among the retained regions, and a clustering task with the number of groups estimated by the silhouette; the repository also holds a threshold-rule study and a sensitivity analysis, to the bound L_{UB}, of an eligibility rule that an earlier draft defined and this paper does not. None of these is reported here, because the paper compares distribution-level measures with the single-summary SMD and prescribes no grouping, threshold or eligibility procedure; their full outputs are in the repository (Data availability statement).

The measures are scored against the construction label on a selection task. An anchor and nine candidate regions are drawn; each measure ranks the candidates by their distance to the anchor and retains the k closest, where k is the number of true matches. Giving every measure the same k removes threshold choice from the comparison, though supplying k is itself an oracle that the application does not use (Section 4). Performance is summarized by the AUC for discriminating true matches from discordant candidates (chance = 0.5) and by the probability that the k retained regions include at least one discordant region. The task ran 10,000 replications per cell at the base grid (n = 25–100 per region) and 3,000 at the extended grid (n = 150–2,000). The extended grid, the accuracy target and the definitions of the two behaviours below were fixed after the base-grid results had been seen, and the Monte Carlo intervals of Table 1 were added later; the reported target is the middle of the three examined.

3.1.2 | Two Behaviours: Blind and Recovering

Extending the sample-size grid to n = 2,000 separates the measures into two behaviours (Figure 2). A measure is blind in a cell when its AUC stays at chance through n = 2,000: the population value of its distance carries no information about the construction label, so the task is unsolvable for that measure at any sample size. This is an identification failure, not a power failure. It is exact by construction in every mean-matched cell: the population SMD is zero there, whatever the sample size. On the selection task, SMD is blind in 7 of the 13 cells, counting each world's combined cell: every cell of Set 3, the two mean-matched cells of Set 4, the scale cell of Set 1, and the shape cell of Set 2 (AUC 0.49–0.51 across the full grid in all of them).

A measure is recovering when its AUC reaches the target and stays above it at every larger sample size: identification succeeds, and what remains is a required-sample-size question. W_1 and the KS statistic are recovering in every cell. In particular, the KS statistic is blind nowhere: in the Set 4 cell where it is weakest, asymmetric severity, it still reaches selection AUC 0.98 at n = 2,000 (0.99 in the symmetric-severity cell).

3.1.3 | Selection: Required Sample Sizes and Error Rates

Table 1 reports, for all 13 cells, the per-region sample size at which each measure's selection AUC reaches 0.9, with 95% Monte Carlo intervals. In the symmetric-severity cell, for example, W_1 reaches the target at n = 42 and the KS statistic at n = 1,047, a 25-fold difference on the same data-generating process, and W_1's own requirement ranges across the task from 42 (symmetric severity) to 690 (shape-symmetric mixtures) in Sets 3 and 4. At the AUC 0.9 target the 95% intervals place W_1 below the KS statistic in 11 cells—three of them the Set 4 subgroup cells, where the advantage follows from the construction (Section 3.1.1), and among the others all six cells of Sets 1 and 2, which contain no outlying subgroup and where neither measure is favoured by

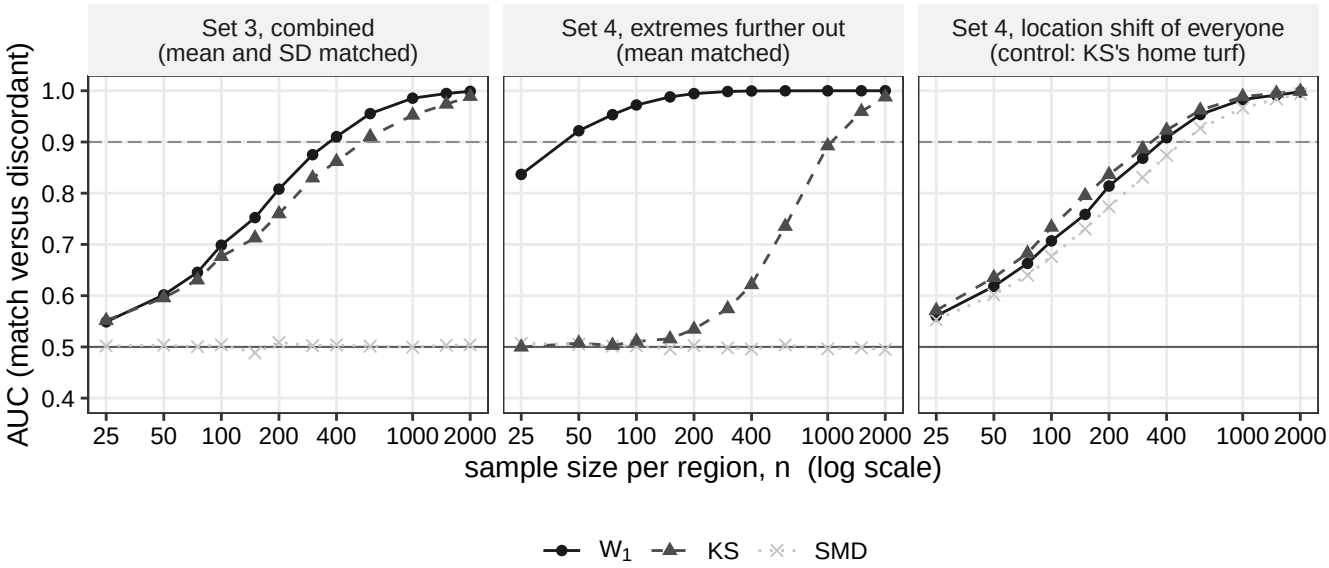


FIGURE 2 Selection AUC (true matches versus discordant candidates) as a function of per-region sample size (n = 25–2,000, log scale) for three distance measures in three cells of Study 1: the Set 3 combined cell (mean- and SD-matched mixtures), the Set 4 symmetric-severity cell (mean-matched displaced extremes), and the Set 4 bulk-shift control cell (uniform location shift of the whole population). The dashed horizontal line marks AUC = 0.9; chance corresponds to 0.5.

construction—place the KS statistic below W_1 in 1, the bulk-shift control, and overlap in 1, the shape-symmetric mixture (690 [661, 718] against 716 [687, 747]). In the bulk-shift control the KS statistic reaches the target first (333 versus 377), whereas in the location cells of Sets 1 and 2, which contain no outlying subgroup, W_1 does (33 versus 39; 129 versus 157). A check registered after an exploratory look (protocol, with the prior look disclosed, and outputs in the repository) compares the two anchors: for the same two-unit shift at n = 200 per region, the Set 4 anchor population multiplies the null sampling standard deviation of $\widehat{W}_1$ by 1.62 (Monte Carlo SE 0.04) relative to $N(50, 10^2)$ while leaving that of the distribution-free KS statistic unchanged (1.00, SE 0.02), and the paired AUC difference between W_1 and the KS statistic changes sign, from 0.029 (SE 0.002) under the Gaussian anchor to -0.032 (SE 0.004) under the Set 4 anchor; the control is reported as a loss for W_1. In the shape-symmetric mixture the two AUC curves cross near the target, the KS statistic discriminating better at the smallest sample sizes and W_1 at the largest, so the overlap marks the crossing rather than equal performance; the order in that cell depends on the target, W_1 reaching AUC 0.95 first with separated intervals, while at AUC 0.8 the point estimates favour the KS statistic and the intervals overlap. Where SMD is not blind, it reaches the target with sample sizes close to W_1's in the Gaussian and log-normal location cells (Set 1: 30 against 33; Set 2: 134 against 129), and with more in the combined cells (67 against 35; 529 against 60) and in the two Set 4 cells where it is not blind (Table 1).

The required-n comparison is complemented by the error rate at a fixed sample size. At n = 100 in Set 4 (all contrast types pooled), the probability that the three retained regions include at least one discordant region is 0.718 for W_1 and 0.971 for the KS statistic (chance 0.988); three of the four discordant types in this pool favour W_1 by construction (Section 3.1.1).

3.2 | Study 2: Estimation Properties

3.2.1 | Design

We designed seven systematic scenarios for methodological validation, examining controlled distributional differences in location, scale, scale combined with skewness (S6, a mean-matched log-normal), and a location–scale combination. Table 2 summarizes these scenarios with their true W_1 values, in the original units of the effect modifier.

For each scenario, we generated samples of size n = 50, 100, and 200 per region. We performed 10,000 replications per scenario–sample size combination; the Monte Carlo standard error is at most 0.005 for any reported proportion, and over the 21

TABLE 1 Per-region sample size at which selection AUC reaches 0.9, by cell and measure, interpolated on $\log n$ over the grid n = 25–2,000, with 95% Monte Carlo intervals from the delta method applied to the log-n interpolation, using the Monte Carlo standard errors of the two AUC estimates that bracket the crossing. Cells whose AUC stays at chance (0.49–0.51) across the full grid are marked "blind"; ≤ 25 marks a target reached at the smallest grid size, where no interval is available, and such a cell counts as separated when 25 lies below the other measure's lower limit.

Cell	W_1 [95% MC interval]	KS [95% MC interval]	SMD [95% MC interval]	W_1 vs KS
Set 1, location	33 [31, 33]	39 [37, 40]	30 [29, 31]	separate (W_1 fewer)
Set 1, scale	≤ 25	76 [73, 78]	blind	separate (W_1 fewer)
Set 1, combined	35 [33, 36]	62 [60, 63]	67 [64, 68]	separate (W_1 fewer)
Set 2, location	129 [123, 135]	157 [149, 165]	134 [127, 140]	separate (W_1 fewer)
Set 2, shape	32 [30, 33]	77 [75, 79]	blind	separate (W_1 fewer)
Set 2, combined	60 [58, 62]	188 [182, 193]	529 [509, 548]	separate (W_1 fewer)
Set 4, symmetric severity	42 [41, 43]	1,047 [1,011, 1,083]	blind	separate (W_1 fewer)
Set 4, symmetric prevalence	98 [94, 101]	367 [356, 378]	blind	separate (W_1 fewer)
Set 4, asymmetric severity	67 [64, 69]	1,252 [1,226, 1,278]	530 [507, 554]	separate (W_1 fewer)
Set 4, bulk shift (control)	377 [362, 392]	333 [318, 348]	489 [469, 509]	separate (KS fewer)
Set 3, combined	368 [354, 381]	550 [528, 572]	blind	separate (W_1 fewer)
Set 3, shape-skewed	537 [520, 554]	905 [865, 945]	blind	separate (W_1 fewer)
Set 3, shape-symmetric	690 [661, 718]	716 [687, 747]	blind	overlap

Cells are contrast types within each world; the "combined" cell holds discordant regions that differ from the anchor in both of the world's contrast types at once (in Set 3, asymmetric and bimodal), not a pool of the other cells.
Intervals reflect only the Monte Carlo error of the two AUC estimates that bracket the crossing, not the error of interpolating between grid points; limits are rounded outward to whole patients. "Separate" means the W_1 and KS intervals do not overlap. All measures are computed on the same replications, so non-overlap is a conservative indication of a difference and overlap does not show equal performance.

TABLE 2 Study 2 simulation scenarios with true W_1 values.

ID	Description	Distribution 1	Distribution 2	True W_1
S1	Null (identical)	$N(50, 10^2)$	$N(50, 10^2)$	0.00
S2	Location 0.2σ	$N(50, 10^2)$	$N(52, 10^2)$	2.00
S3	Location 0.5σ	$N(50, 10^2)$	$N(55, 10^2)$	5.00
S4	Location 1.0σ	$N(50, 10^2)$	$N(60, 10^2)$	10.00
S5	Scale 1.5×	$N(50, 10^2)$	$N(50, 15^2)$	3.99
S6	Skew + scale (Log-normal)	$N(50, 10^2)$	$\text{LogN}(3.787, 0.5^2)$	12.16
S7	Location + Scale	$N(50, 10^2)$	$N(55, 15^2)$	5.83

The Log-normal in S6 has $\sigma_{\log} = 0.5$ and $\mu_{\log} = \log 50 - \sigma_{\log}^2/2 = 3.787$, giving mean 50, SD 26.6 (against 10 for Distribution 1, so the contrast is in scale as well as shape), CV = 53.3% and skewness 1.75. S7 combines location and scale differences. True W_1 values for S1–S4 are exact ($W_1 = |\mu_1 - \mu_2|$ for equal-variance normals); S5 uses the closed form $W_1 = \sqrt{2/\pi}\,|\sigma_1 - \sigma_2|$; S6 and S7 are obtained by numerical quadrature of $W_1 = \int |F_1(t) - F_2(t)|\,\mathrm{d}t$, with the S6 integral split at $t = 0$ to accommodate the log-normal support. The table shows true W_1 to two decimals; the performance measures use the exact values 3.989423 (S5), 12.159806 (S6) and 5.833155 (S7), and the share of intervals lying above the truth in Section 3.2.2 was computed with Monte Carlo truths for S6 and S7, which does not change it at the precision reported.

cells it is at most 0.026 for bias, 0.021 for RMSE and 0.024 for mean CI width. Every replication produced a finite estimate and interval. Bootstrap confidence intervals were computed as in Section 2.3 with B = 2,000 resamples. Study 2 was run in R with the bootstrap in C++ via Rcpp, replications distributed over a PSOCK cluster with L'Ecuyer–CMRG random-number streams seeded per cell (`clusterSetRNGStream`, seed 42 + 1,000k for cell $k = 1, \ldots, 21$, scenarios in order and n within scenario); because replicates are assigned to workers dynamically, individual replicates are not bit-reproducible, and the summaries are reproducible to Monte Carlo error.

We evaluated the operating characteristics of the sample estimator $\widehat{W}_1$ for the population value W_1, in the original units of the effect modifier:

1. Bias: $\text{Mean}(\widehat{W}_1) - W_1$
2. Root mean squared error (RMSE): $\sqrt{\text{Mean}[(\widehat{W}_1 - W_1)^2]}$
3. Coverage probability: Proportion of 95% bootstrap CIs $[L, U]$ with $L \leq W_1 \leq U$. Under S1 ($W_1 = 0$) coverage is structurally zero (Section 3.2.2)
4. CI width: Mean width of bootstrap CIs, reflecting estimation precision

3.2.2 | Results

Table 3 presents bias, coverage, RMSE and mean CI width of $\widehat{W}_1$ across scenarios and sample sizes, in the original units of the effect modifier (scenarios are specified with $\sigma = 10$). The estimator showed positive bias under the null scenario (S1), decreasing from 2.46 at $n = 50$ to 1.26 at $n = 200$: the finite-sample bias of Section 2.3 ($E\,\widehat{W}_1 \geq W_1$), at its largest when the true value equals zero ($F_1 = F_2$ almost everywhere). The estimator remains consistent ($\widehat{W}_1 \to 0$ as $n \to \infty$ under the null), and the bias fell at a rate consistent with $n^{-1/2}$: at $n = 200$ it is about half that at $n = 50$.

TABLE 3 Bias, coverage of the 95% percentile bootstrap confidence interval, RMSE and mean CI width of $\widehat{W}_1$ by scenario and sample size, in the original units of the effect modifier.

Scenario	True W_1	n	Bias	Coverage	RMSE	Mean CI width
S1 (Null)	0.00	50	2.463	0.000	2.63	4.19
		100	1.759	0.000	1.87	2.94
		200	1.260	0.000	1.34	2.07
S2 (0.2σ)	2.00	50	0.989	0.703	1.60	4.67
		100	0.472	0.904	1.13	3.57
		200	0.174	0.947	0.85	2.84
S3 (0.5σ)	5.00	50	0.175	0.951	1.83	6.22
		100	0.051	0.951	1.37	4.96
		200	0.021	0.948	0.98	3.77
S4 (1.0σ)	10.00	50	0.034	0.942	1.98	7.52
		100	−0.013	0.945	1.42	5.45
		200	0.005	0.948	1.00	3.89
S5 (Scale)	3.99	50	0.830	0.862	1.64	5.65
		100	0.393	0.917	1.11	4.10
		200	0.187	0.946	0.77	2.97
S6 (Skew)	12.16	50	0.377	0.945	2.61	9.73
		100	0.203	0.942	1.86	7.04
		200	0.111	0.947	1.29	5.03
S7 (Loc+Scale)	5.83	50	0.638	0.928	2.00	6.97
		100	0.340	0.935	1.45	5.27
		200	0.193	0.943	1.01	3.87

Bias is $\text{Mean}(\widehat{W}_1) - W_1$ and RMSE is $\sqrt{\text{Mean}[(\widehat{W}_1 - W_1)^2]}$; coverage is the proportion of the 95% percentile bootstrap intervals that contain W_1, and is zero under S1 because $W_1 = 0$ lies at the parameter space boundary and a percentile interval cannot contain it; CI width is the mean of (upper − lower). Scenarios are specified with $\sigma = 10$. Results based on 10,000 replications with $B = 2{,}000$ bootstrap resamples.

For the non-null scenarios away from the boundary (S3–S7), bias was small relative to each scenario's true W_1 and diminished with sample size: at $n = 200$ it was at most 0.193 (S7) (Figure 3(A)), and in S4 it was within Monte Carlo error at every n. Near the boundary (S1, S2) the positive bias was larger relative to the true value, as expected from the upward bias of Section 2.3. Under the null (S1), where the limit of $\widehat{W}_1$ is not centred, the bias fell at the $n^{-1/2}$ rate; in the other non-null scenarios (S2, S3, S5–S7; S4's bias is within Monte Carlo error) it fell faster, the bias at $n = 200$ being at most one third of that at $n = 50$ (Table 3), as expected away from the boundary, where the limit of $\widehat{W}_1 - W_1$ is centred.

For the non-null scenarios S3–S7 at $n \geq 100$ (Table 3), coverage ranged over 0.92–0.95; of these ten cells, 5 lie more than two Monte Carlo standard errors below 0.95, the lowest being S5 at $n = 100$ (0.917). The shortfall is largest in S5 and S7, whose CDFs cross; there the intervals lie above the true value, in line with the upward bias of Section 2.3, which relative to the sampling standard deviation of $\widehat{W}_1$ is 0.38 and 0.24 at $n = 100$ against 0.04 for S3. In S6 (0.11), whose bias is smaller relative to its sampling standard deviation than in S5 and S7, and in S4 at $n = 100$, whose bias is within Monte Carlo error, the shortfall is small and the misses fall on both sides (Table 4), so a given n does not by itself fix the coverage. S4 (1.0σ location shift) showed coverage of 0.942–0.948 across sample sizes, with no degradation at larger n, within two Monte Carlo standard errors of 0.95 only at $n = 200$. S5 (scale) and S2 (small location shift, near the boundary) showed undercoverage at $n = 50$ (0.862 and 0.703) that improved to 0.946 and 0.947 at $n = 200$.

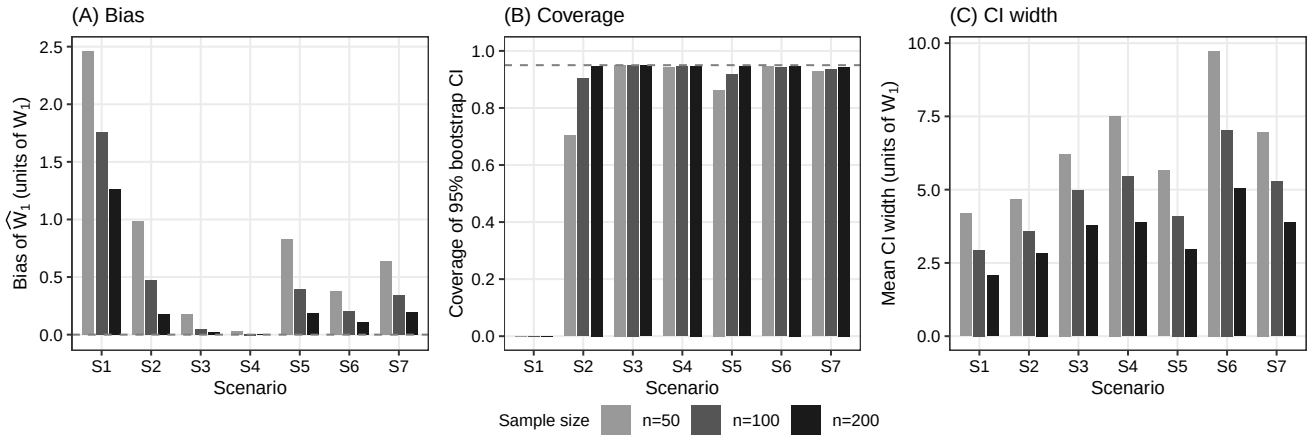


FIGURE 3 Estimation properties of the sample Wasserstein-1 estimator $\widehat{W}_1$, in the original units of the effect modifier, across scenarios (S1–S7) and sample sizes ($n = 50, 100, 200$). Panel (A): bias, with dashed horizontal line at zero. Panel (B): coverage probability of 95% percentile bootstrap CIs, with dashed line at the nominal 0.95 level; S1 coverage is structurally zero because the true value lies at the parameter space boundary (see Section 3.2.2). Panel (C): mean CI width in units of W_1.

Coverage for S1 (null) is zero (Table 3, Figure 3(B)) because its true W_1 lies at the parameter space boundary. This does not indicate estimator failure—the estimator is consistent—but a percentile interval cannot contain a true value of zero at any sample size, because with continuous data every bootstrap replicate of $\widehat{W}_1$ is positive; whether an observed distance is distinguishable from no difference is instead read against the null floor. Where coverage falls short in the non-null scenarios (S2, S5 and S7 at smaller n), the interval misses from below: the true value exceeded the upper limit in at most 1.1% of replications in S2, S5 and S7 at any sample size (never in S2), against 2.5% nominal. The upper confidence limit is therefore a conservative reading of how large a difference could be in S2, S5 and S7; this is not a property of every scenario whose CDFs cross, since in S6 the true value exceeded the upper limit at about the nominal rate.

Monte Carlo standard errors for the reported coverage probabilities are at most 0.005 at 10,000 replications (largest observed 0.0046). A difference between two cells is within simulation noise when it is smaller than about 2.8 times the standard error, and a single cell is consistent with nominal coverage when it lies within about twice its own standard error of 0.95. Because the application reads an unresolved distance at its upper limit, the one-sided coverage of the 97.5% upper limit, $P(W_1 \leq U)$, is the operating characteristic that matters there: for S3–S7 at $n \geq 100$ it ranged over 0.9710–0.9999 against a nominal 0.975: above nominal where the interval lies above the truth (S5, S7), near nominal in S3 and S6, and below it in S4 at $n = 100$ (Table 4). The basic (reflected) interval $[\max(0, 2\widehat{W}_1 - U), \max(0, 2\widehat{W}_1 - L)]$, which shifts the percentile interval by the bootstrap estimate of the bias, covered less than the percentile interval in every non-null cell from $n = 100$ (0.870–0.948 in S3–S7) but more near the boundary (under S1 its coverage, 0.867 at $n = 50$, equals the share of replications whose lower limit is truncated at zero; in S2 at $n = 50$ it was 0.809 against 0.703), so this bias correction trades coverage away from the boundary for coverage at it (Table 4); these comparisons were computed after the simulation from the stored replications and were not pre-specified.

A registered extension of Study 2 (protocol and outputs in the repository; its first eight cells were registered after the external review and before the run, and three further cells—the two at the application's operating point and one more integer-grid cell at $n = 1{,}000$—as an amendment registered before those cells ran) added cells outside the original design: unequal regional sizes, a heavy-tailed reference distribution, integer-grid data, $n = 1{,}000$ per region and the application's operating point (the cells are described in the note to Table 5). Across the ten non-null cells coverage ranged over 0.921–0.952 (Monte Carlo standard error at most 0.005; 2000–10,000 replications per cell), lowest for the shift on the t_3 reference (0.921, with a positive bias of 0.49); at the application's operating point it was 0.951 with equal sizes and 0.947 with unequal, rounded samples, with upper-limit coverage 0.9825 and 0.9775 (Table 5).

RMSE decreased consistently with increasing n (Table 3), at the $O(n^{-1/2})$ rate expected from Wasserstein plug-in theory; for S3, for example, it fell from 1.83 at $n = 50$ to 0.98 at $n = 200$. Mean CI width narrowed with n (Figure 3(C)), with CIs at $n = 100$ for the non-null scenarios spanning roughly 3.6–7.0 in the units of the effect modifier. For S3 on the age scale with the illustrative bound $L_{\mathrm{UB,age}} = 0.01$/yr of the GUSTO-I application, a $\widehat{W}_1$ CI width of 4.96 years at $n = 100$ corresponds to a $\Delta_{\max}$

TABLE 4 Coverage of the 95% percentile interval, of its one-sided 97.5% upper limit, and of the basic (reflected) interval, with mean widths, by scenario and sample size, in the original units of the effect modifier.

Scenario	n	Percentile coverage	Upper-limit coverage	Basic coverage	Mean width (percentile)	Mean width (basic)
S1 (Null)	50	0.000	1.0000	0.867	4.19	3.08
	100	0.000	1.0000	0.868	2.94	2.15
	200	0.000	1.0000	0.872	2.07	1.52
S2 (0.2σ)	50	0.703	1.0000	0.809	4.67	3.82
	100	0.904	1.0000	0.737	3.57	3.12
	200	0.947	1.0000	0.740	2.84	2.65
S3 (0.5σ)	50	0.951	0.9898	0.788	6.22	6.02
	100	0.951	0.9815	0.870	4.96	4.95
	200	0.948	0.9764	0.924	3.77	3.77
S4 (1.0σ)	50	0.942	0.9729	0.925	7.52	7.52
	100	0.945	0.9710	0.940	5.45	5.45
	200	0.948	0.9760	0.946	3.89	3.89
S5 (Scale)	50	0.862	1.0000	0.866	5.65	5.40
	100	0.917	0.9999	0.886	4.10	4.08
	200	0.946	0.9977	0.922	2.97	2.97
S6 (Skew)	50	0.945	0.9796	0.919	9.73	9.73
	100	0.942	0.9735	0.934	7.04	7.04
	200	0.947	0.9744	0.948	5.03	5.03
S7 (Loc+Scale)	50	0.928	0.9991	0.859	6.97	6.89
	100	0.935	0.9927	0.897	5.27	5.27
	200	0.943	0.9888	0.931	3.87	3.87

Upper-limit coverage is $P(W_1 \leq U)$; under S1 it is identically 1 because $U > 0 = W_1$. The basic interval is $[\max(0, 2\widehat{W}_1 - U), \max(0, 2\widehat{W}_1 - L)]$; under S1 its coverage equals the share of replications whose lower limit is truncated at zero. Computed from the same 10,000 replications as Table 3; not pre-specified.

TABLE 5 Registered extension of Study 2: unequal regional sizes, a t_3 reference distribution, integer-grid data, n = 1,000 per region and the application's operating point (0.2 SD). Bias, RMSE and mean width in the original units; coverage of the 95% percentile interval and of its one-sided 97.5% upper limit.

Cell	n_1	n_2	Reps	True W_1	Bias	RMSE	Coverage	Upper-limit coverage	Mean width
E1 unequal n	100	300	10,000	5.00	0.056	1.14	0.947	0.9784	4.25
E1 unequal n	200	1,000	5,000	5.00	0.017	0.78	0.941	0.9734	2.97
E2 heavy tails: t_3, null	100	100	10,000	0.00	1.958	2.09	0.000	1.0000	3.08
E2 heavy tails: t_3, shift	100	100	10,000	5.00	0.491	1.28	0.921	0.9947	4.44
E3 integer grid	100	100	10,000	5.00	0.064	1.38	0.945	0.9779	4.97
E3 integer grid	200	200	10,000	5.00	0.021	0.98	0.950	0.9789	3.77
E3 integer grid	1,000	1,000	2,000	5.00	0.019	0.45	0.952	0.9785	1.74
E4 large n: S3	1,000	1,000	5,000	5.00	−0.002	0.45	0.946	0.9746	1.74
E4 large n: S5	1,000	1,000	5,000	3.99	0.034	0.35	0.944	0.9826	1.35
E5 small distance	1,000	1,000	2,000	2.00	0.034	0.44	0.951	0.9825	1.66
E6 small distance, unequal n, rounded	2,900	1,200	2,000	2.00	−0.001	0.35	0.947	0.9775	1.37

Cells: E1, unequal regional sizes (100 against 300; 200 against 1,000); E2, a t_3 reference scaled to SD 10, under the null and under a 5-unit shift; E3, integer-grid data (the S3 contrast rounded to integers at n = 100, 200 and 1,000); E4, n = 1,000 per region for the S3 and S5 contrasts; E5 and E6, the application's operating point (the 0.2 SD contrast of S2 at n = 1,000 per region, and at 2,900 against 1,200 with both samples rounded to multiples of 10, as systolic blood pressure is recorded).
Under the t_3 null the true W_1 is zero, so percentile coverage is structurally zero and upper-limit coverage identically 1, as under S1. The rounded-data truth is the W_1 of the rounded population distributions. Replications per cell as shown; Monte Carlo standard error of coverage at most 0.005. Protocol `PROTOCOL_study2_extension.md`, registered after the external review; the two operating-point cells and the integer-grid cell at n = 1,000 were added as an amendment before they were run.

interval about 5 times the 1%pt margin used in Section 4.4, so n = 100 per region does not by itself make the $\Delta_{\max}$ interval narrow relative to a clinical margin.

Under the null scenario (S1), the informative summary is the upper tail rather than the mean: the 95th percentile of $\widehat{W}_1$ under S1 is 4.21, 2.97 and 2.11 at n = 50, 100 and 200 for this scenario (σ = 10, equal n), and an observed distance at or below the corresponding value is not evidence of a difference.

4 | APPLICATION: HYPOTHETICAL THROMBOLYTIC MRCT USING GUSTO-I DATA

We illustrate the per-effect-modifier Wasserstein-1 framework through a hypothetical scenario grounded in publicly available data from GUSTO-I, a large international trial in acute myocardial infarction (AMI). A sponsor developing a novel thrombolytic agent (Drug T) for AMI plans a Phase 3 MRCT across multiple regions and uses IPD from GUSTO-I (N = 40,830; 16 regions)[28] to quantify how the distributions of candidate effect modifiers in other regions differ from those in a designated anchor region, treating the GUSTO-I regions as stand-ins for the regional populations the new trial would enrol. GUSTO-I was not designed as an MRCT and its regions are anonymized.

4.1 | Scenario Design and Effect Modifier Selection

For this illustrative exercise, the sponsor takes age and systolic blood pressure (SBP) as the candidate effect modifiers for Drug T, suggested by exploratory Phase 2 analyses and biological plausibility whose precision is insufficient for numerical estimation of L_{clinical}. Class-level indirect evidence can at most inform a sponsor-specified upper bound (Section 2.4): the Fibrinolytic Therapy Trialists' (FTT) collaborative meta-analysis of existing fibrinolytic agents[26] concluded that, among patients presenting with ST elevation or bundle-branch block, benefit is present irrespective of age and of blood pressure, and reports no per-unit CATE slope on the absolute mortality scale for either, so class transferability would not yield a defensible numerical prior for Drug T. We therefore treat L_{clinical} as unknown a priori for both candidate effect modifiers, and apply the L^* reverse-calculation pathway (equation 10) to both.

The sponsor selects Region 8 as the anchor region, representing for instance a regulatory market where local efficacy evidence is required; it comprises n = 2,916 patients (Table 6), and the anchor role, not its size, is what the exercise illustrates. The GUSTO-I records are used as recorded: 6 patients have a systolic blood pressure of 0 mmHg and 3 are older than 100 years, and W_1 weights each observation by its displacement. Excluding these records changes $\widehat{W}_1$ by at most 0.02 years and 0.07 mmHg (to two decimals), exchanges 2 adjacent SBP ranks and changes no null-floor classification below; winsorizing both variables at the 1st and 99th percentiles moves $\widehat{W}_{1,\text{SBP}}$ by up to 0.63 mmHg and changes 4 of the 30 ranks; these checks were made after the main analysis. Systolic blood pressure is also heaped: the share of readings on a multiple of 10 mmHg ranges from 19.9% to 79.5% across the 16 regions and is 79.5% in Region 8, so differences in recording practice may contribute to the SBP distances (not quantified here) and the step CDFs of Appendix A.2 apply.

TABLE 6 Region 8 baseline characteristics (GUSTO-I).

Candidate effect modifier	Mean	SD	Skewness
Age (years)	60.2	12.1	−0.17
SBP (mmHg)	132.4	22.9	0.20

n = 2,916.

4.2 | Null Floor: Which Distances Are Distinguishable from No Difference?

Before interpreting the distances of Section 4.3, we computed the null floor of Section 2.3 separately for each candidate partner, from two independent resamples, with replacement, of Region 8's (R8) empirical distribution at sizes n_{R8} = 2,916 and n_{partner}, recomputing $\widehat{W}_1$ over 2,000 repetitions. We call an observed distance unresolved when it lies at or below the 95th percentile of its null distribution (a one-sided test of $F_1 = F_2$ at level 0.05, applied separately to each pair). A permutation test of $F_1 = F_2$ on the pooled sample (2,000 permutations), which is exact under exchangeability, classifies all 30 distances in the same way (30 of 30 agree; 19 fall below a Bonferroni-adjusted 0.05/30), and drawing the null from the partner's own distribution instead of the anchor's leaves every distance on the same side of its 95th percentile, so the anchor-resampled floor is retained as the display reference. Table 7 summarises the result.

The two candidate effect modifiers differ in how many distances clear the null floor, and the difference matters for how the subsequent distances should be read. On age, the distances of six of the fifteen partners—R1, R4, R5, R7, R9 and R15—are

TABLE 7 Null floor of $\widehat{W}_1$ across the 15 candidate partners, by effect modifier. Ranges are across partners; the floor depends on each partner's sample size. Unresolved distances are those whose observed $\widehat{W}_1$ falls at or below the 95th percentile of the null distribution.

Candidate effect modifier	Null 95th pct.	Unresolved
Age (years)	0.618–0.900	6 / 15
Systolic blood pressure (mmHg)	1.176–1.691	1 / 15

unresolved at level 0.05: they are ones the estimator would produce with appreciable probability from identical distributions. On systolic blood pressure, by contrast, fourteen of fifteen are resolved, the single exception being R2.

The small age distances of the six partners unresolved at level 0.05 are therefore not evidence of a difference; how large their differences could be is read from the upper confidence limits in Table 8, which for small distances near the boundary err above rather than below the true value in the Study 2 scenarios (Section 3.2.2). Neither resolution against the null floor nor the coverage of the percentile interval is fixed by the sample size alone; both depend on the distance relative to the sampling variability of $\widehat{W}_1$ (Section 3.2.2): the partner samples range from 1,231 to 4,352 observations, but for a distance at or below the null floor the interval is read at its upper limit whatever the sample size. The GUSTO-I distances are 0.03–0.30 SD of Region 8's distributions, below the smallest non-null distance of the original Study 2 design; the extension cells at this operating point (Table 5) gave percentile coverage 0.951 and 0.947 and upper-limit coverage 0.9825 and 0.9775, so the intervals of Table 8 are read with that evidence rather than with the original cells.

4.3 | Distributional Assessment: Per-Effect-Modifier W_1 Across 15 Partner Regions

Study 1 showed that W_1 and the KS statistic both reach selection AUC 0.9 in every simulated world (Section 3.1.3). Study 1's required sample sizes refer to its simulated contrasts and do not transfer to the GUSTO-I distances, which are read against their null floors (Section 4.2); W_1 is used because its heterogeneity bound has a per-unit clinical slope as its constant (Section 2.2). We therefore computed $\widehat{W}_1$ between Region 8 and each of the remaining 15 regions for both candidate effect modifiers and ordered the partners by it, expressed in the original measurement units (years for age, mmHg for SBP), with percentile bootstrap 95% confidence intervals (B = 2,000). Table 8 and Figure 4 (a forest plot) present the results.

TABLE 8 Per-effect-modifier Wasserstein-1 distance $\widehat{W}_1$ in original variable units (years for age, mmHg for SBP) with 95% percentile bootstrap CIs for Region 8 versus each partner region (GUSTO-I), listed by region number, with each partner's rank by $\widehat{W}_1$ within each candidate effect modifier.

Partner	n	$\widehat{W}_{1,\text{age}}$ (yr) [95% CI]	Rank	$\widehat{W}_{1,\text{SBP}}$ (mmHg) [95% CI]	Rank
R1	2188	0.65 [0.51, 1.03]	1–6† [2, 10]	3.61 [2.68, 4.93]	7 [3, 10]
R2	2952	2.15 [1.57, 2.76]	14 [13, 15]	0.95 [0.76, 1.82]	1† [1, 2]
R3	2030	2.61 [1.96, 3.28]	15 [14, 15]	3.35 [2.32, 4.65]	4 [2, 9]
R4	2876	0.56 [0.37, 1.16]	1–6† [1, 9]	2.60 [1.82, 3.75]	3 [2, 6]
R5	1909	0.39 [0.35, 0.92]	1–6† [1, 9]	3.37 [2.32, 4.59]	6 [3, 9]
R6	1585	0.91 [0.63, 1.39]	9 [4, 11]	3.37 [2.06, 4.77]	5 [2, 10]
R7	3150	0.40 [0.30, 0.95]	1–6† [1, 9]	5.07 [4.03, 6.22]	12 [9, 12]
R9	3123	0.59 [0.40, 1.12]	1–6† [1, 9]	6.80 [5.71, 8.01]	15 [13, 15]
R10	1717	1.38 [0.79, 2.13]	12 [4, 14]	4.00 [2.88, 5.39]	8 [4, 11]
R11	2491	1.27 [0.76, 1.91]	11 [7, 13]	6.44 [5.25, 7.71]	14 [12, 15]
R12	4352	0.86 [0.47, 1.43]	8 [2, 11]	6.40 [5.39, 7.46]	13 [13, 15]
R13	2297	1.15 [0.76, 1.74]	10 [7, 12]	2.40 [1.42, 3.72]	2 [1, 5]
R14	3437	0.70 [0.42, 1.29]	7 [1, 10]	4.36 [3.41, 5.44]	10 [6, 12]
R15	2576	0.61 [0.40, 1.25]	1–6† [1, 10]	4.72 [3.60, 5.94]	11 [7, 12]
R16	1231	1.74 [1.16, 2.47]	13 [11, 14]	4.28 [3.06, 5.71]	9 [3, 13]

Point estimates with 95% percentile bootstrap CIs (B = 2,000, seed = 2026). Regions are listed by region number; Rank orders the partners by $\widehat{W}_1$ within each candidate effect modifier (1 = closest to Region 8); the bracketed rank interval is the 2.5th–97.5th percentile of the rank over 1,000 patient-level bootstrap replicates that resample within every region (Region 8 shared across partners; seed 20261011), and adjacent ranks whose intervals overlap are not resolved against each other. †Distances unresolved against their null floor (Table 7) are not ordered among themselves and share the point-rank range shown; the bracketed interval that follows it is that partner's own bootstrap rank interval, which for these partners reaches the ninth or tenth position. R6 and R5, adjacent in the SBP ranking, differ in $\widehat{W}_{1,\text{SBP}}$ by 0.003 mmHg, so their ranks are effectively tied.

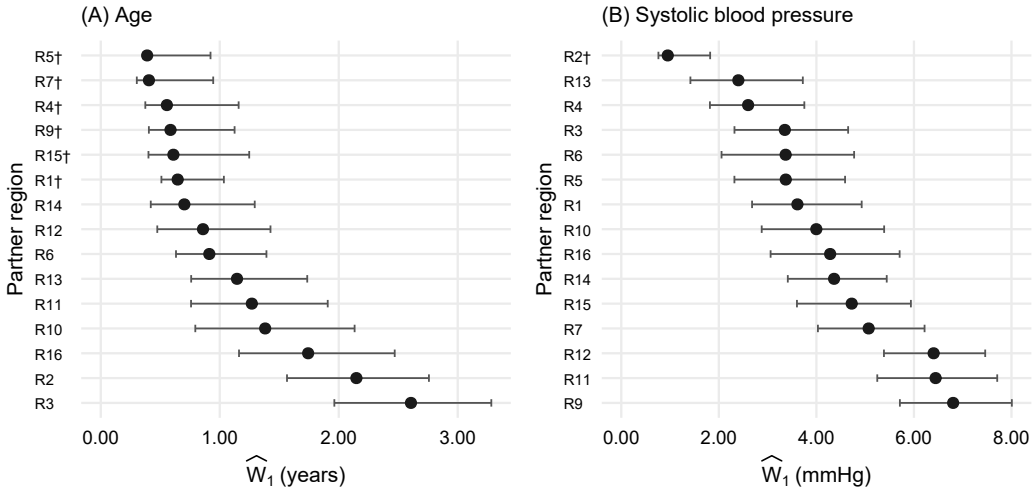


FIGURE 4 Forest plots of $\widehat{W}_1$ point estimates in original units (years for age, panel A; mmHg for SBP, panel B) with 95% percentile bootstrap CIs. Partners are ordered by ascending $\widehat{W}_1$ within each panel; ordering and horizontal scale differ between panels. †Distance unresolved against its null floor (Table 7).

Several partners fall in the middle of each ranking where uncertainty intervals overlap substantially, and the rank intervals of Table 8, obtained by resampling patients within every region, show how far such a partner can move: a mid-table rank is not resolved against its neighbours (R6 ranks ninth on age and fifth on SBP, and both of its rank intervals span most of the middle of the table), whereas the most distant partners, R2 on age and R9 on SBP, have rank intervals confined to the most distant positions, so a bare rank is not read as a resolved order.

A contrast emerges between R2 and R9. R2 has the second-largest $\widehat{W}_{1,\text{age}}$ (2.15 years) but the smallest $\widehat{W}_{1,\text{SBP}}$ (0.95 mmHg), while R9 is among the six smallest $\widehat{W}_{1,\text{age}}$ (0.59 years) but the largest $\widehat{W}_{1,\text{SBP}}$ (6.80 mmHg). Both small distances are unresolved against their null floors (Section 4.2), so the reversal rests on the two large ones. The reversal illustrates that the distance on one candidate effect modifier need not carry information about the distance on another; Section 5 reports the rank agreement across the 15 partners.

4.4 | Clinical Interpretation

Table 9 reports, for each partner region, L^* with its percentile interval (Section 2.4). Each interval is read against the illustrative bound L_{UB} for its candidate effect modifier. Where the whole interval lies above L_{UB}, even the upper confidence limit of the distributional difference would require a clinical slope above the bound to produce $\Delta_{\text{clin}} = 1\%\text{pt}$; where the interval contains L_{UB}, the data are compatible with a difference that could and one that could not produce it; where the interval lies below L_{UB}, a slope within the bound could produce it even at the lower confidence limit of the distance. For distances unresolved against the null floor (Section 4.2), only the lower end of the interval, which comes from the upper confidence limit of $\widehat{W}_1$, carries the reading. If the CATE were additionally assumed monotone, equation (7) would replace $\Delta_{\max}$ by the fraction $(W_1 + |\mu_1 - \mu_2|)/(2W_1)$ of it, which ranges from 0.66 to 1.00 across the age distances and from 0.72 to 1.00 across the SBP distances (point estimates, including the unresolved R2 on SBP; the factor would equal 1 only if the partner's CDF never crossed the anchor's), and L^* would be larger by the reciprocal.

5 | DISCUSSION

This paper answers the requirements set out in Section 2.1 and characterizes the estimation of its answer. The W_1 distance meets requirement (i): it resolves distributional differences to which a location summary such as the SMD is structurally blind—in Study 1, SMD is at chance at every sample size examined in 7 of the 13 selection cells, an identification failure that no sample size

TABLE 9 L^* at $\Delta_{\text{clin}} = 1\%$pt with 95% percentile interval for all 15 partner regions, listed by region number. L^* is the minimum clinical slope (per year for age, per mmHg for SBP) required for the observed distributional difference to produce a treatment effect difference of 1%pt on the absolute 30-day mortality scale.

Partner	$\widehat{W}_{1,\text{age}}$ (yr)	$\widehat{W}_{1,\text{SBP}}$ (mmHg)	L^* age (/yr)	L^* SBP (/mmHg)
R1	0.65†	3.61	0.0155 [0.0097, 0.0196]	0.0028 [0.0020, 0.0037]
R2	2.15	0.95†	0.0047 [0.0036, 0.0064]	0.0105 [0.0055, 0.0131]
R3	2.61	3.35	0.0038 [0.0030, 0.0051]	0.0030 [0.0022, 0.0043]
R4	0.56†	2.60	0.0180 [0.0086, 0.0267]	0.0038 [0.0027, 0.0055]
R5	0.39†	3.37	0.0256 [0.0108, 0.0282]	0.0030 [0.0022, 0.0043]
R6	0.91	3.37	0.0110 [0.0072, 0.0158]	0.0030 [0.0021, 0.0049]
R7	0.40†	5.07	0.0247 [0.0106, 0.0330]	0.0020 [0.0016, 0.0025]
R9	0.59†	6.80	0.0171 [0.0089, 0.0247]	0.0015 [0.0012, 0.0018]
R10	1.38	4.00	0.0072 [0.0047, 0.0126]	0.0025 [0.0019, 0.0035]
R11	1.27	6.44	0.0079 [0.0052, 0.0132]	0.0016 [0.0013, 0.0019]
R12	0.86	6.40	0.0116 [0.0070, 0.0211]	0.0016 [0.0013, 0.0019]
R13	1.15	2.40	0.0087 [0.0058, 0.0132]	0.0042 [0.0027, 0.0071]
R14	0.70	4.36	0.0142 [0.0077, 0.0238]	0.0023 [0.0018, 0.0029]
R15	0.61†	4.72	0.0164 [0.0080, 0.0249]	0.0021 [0.0017, 0.0028]
R16	1.74	4.28	0.0057 [0.0040, 0.0086]	0.0023 [0.0018, 0.0033]

Regions are listed by region number; their ranks by $\widehat{W}_1$ are given in Table 8. †Distance unresolved against its null floor (Table 7). The illustrative bounds of Section 2.4 are $L_{\text{UB,age}} = 1 \times 10^{-2}$ /yr and $L_{\text{UB,SBP}} = 2 \times 10^{-3}$ /mmHg; the lower limit of R1's SBP interval, 0.00203, lies just above $L_{\text{UB,SBP}}$.

repairs, whereas W_1 reaches the discrimination target in all four worlds, at $n = 42$–690 per region in Sets 3 and 4 (Section 3.1.2). It meets requirements (ii) and (iii): the heterogeneity bound $\Delta_{\max}$ (equation 9) translates distributional distance into a clinically interpretable scale, is sharp when the slope is the only restriction on the CATE and tightens under monotonicity (equation 7), and its constant—a treatment effect change per unit of the effect modifier—is a clinical slope that the sponsor specifies, informed by clinical evidence. SMD supports no bound valid across the Lipschitz class, since regions with equal means can have $W_1 > 0$ and hence, by equation (5), regional treatment effects differing by up to $L_{\text{clinical}} W_1$. Its estimation is characterized by Study 2: percentile bootstrap intervals whose coverage was 0.92–0.95 for true distances of about 0.4 to 1.2 SD at $n \geq 100$ per region, below 0.95 beyond Monte Carlo error in 5 of the ten cells, with a characterized boundary behaviour near the null (Section 3.2.2).

These findings carry two implications for pooling assessments, illustrated by the GUSTO-I example (Section 4). First, when several candidate effect modifiers are considered, the per-modifier distances need not agree: a partner close to the anchor on one effect modifier can be distant on another, as in the contrast between R2 and R9, so the distance on a modifier left out of the assessment need not be inferable from those included. Second, W_1 ranking alone does not convey clinical weight: within one effect modifier $L^* = \Delta_{\text{clin}}/W_1$ is a strictly decreasing function of W_1, so calibration does not reorder partners; it compares each partner's L^* with the bound L_{UB} specified for that modifier, and both are reported.

These implications also bear on the cluster analysis of regions that Komiyama et al. introduce as one example of forming pooled regions, in which the candidate effect modifiers are combined into a single weighted distance;[6] the clustering task run under our protocol is not reported (Section 3.1.1). Across the 15 partners of Table 8, the orderings by age and by systolic blood pressure distance show no positive agreement (Spearman correlation −0.30; 95% interval −0.51 to 0.02 from the patient-level bootstrap of Table 8; it is computed on the point ordering of all 15 partners, including the distances left unordered by the null floors); the interval excludes the strong positive agreement under which one modifier's distances could stand in for the other's, so a single distance aggregated over the two modifiers could mask differences that a per-modifier assessment retains. How many pooled regions to form, and how orderings that differ across modifiers are reconciled, remain clinical judgments.

Of the limitations set out in Section 2.1, the location-only limitation of SMD is structural and the lack of a clinical scale is addressed by the heterogeneity bound (equation 9), while those of KL divergence are avoided by construction, W_1 being symmetric, finite for any pair of empirical distributions, and computable from CDFs without density estimation; this does not mean returning a different answer in every dataset. In GUSTO-I, whose regional differences are predominantly differences in location (partner standard deviations within a factor of 1.11 of the anchor's), W_1 ordered the 15 partners much as the SMD did (Spearman correlation 0.90, 95% interval 0.70 to 0.99, for age and 0.95, 0.90 to 0.99, for systolic blood pressure, from the patient-level bootstrap of Table 8), and its contribution there was the clinical calibration rather than a different answer; a survey of 1,503 region pairs in GUSTO-I, IST-1 and IST-3 (the first and third International Stroke Trials),[28,31,32] registered after an exploratory scan of the same data had shown the pattern (protocol and outputs in the repository), found agreement of the same order, with

rank correlations between W_1 and |SMD| (pooled SD of the simple-average form) of 0.839–0.996 across 13 trial-by-modifier combinations, five of them in GUSTO-I. The limitations bind where distributional differences go beyond location—mixtures and rare displaced subgroups, configurations that cannot be excluded in advance—and which kind of dataset one holds is not knowable before the distances are computed. The reason to use W_1 is therefore not efficiency against the KS statistic: in the simulated cells W_1 reached the selection target with fewer patients than the KS statistic in 11 of 13 cells at AUC 0.9, three of them by construction, and the KS statistic needed fewer in the location-shift control on the population with contaminated tails (Table 1, Section 3.1.3), while in the three real trials of the survey the two statistics ranked region pairs almost identically (Spearman 0.879–0.989). It is that W_1, like the KS statistic, is not blind where the SMD is, and that it carries a translation to the clinical scale whose constant is a per-unit clinical slope, valid for every CATE in the class and conservative for any particular one.

Because W_1, like other covariate-distribution measures, requires only baseline effect modifier distributions, regional data can be assembled from prior trials, registries, electronic health records, or other real-world evidence (RWE) sources, provided the source population approximates the population each region is expected to enrol. The assessment can therefore be made at the planning stage, before any regional data from the new trial exist.

The W_1 distance is one input within a holistic regulatory evaluation; biological plausibility, clinical relevance, and benefit–risk balance extend beyond what any single distributional measure can quantify.[3,4] The framework has further limitations. It treats continuous effect modifiers marginally, leaving categorical and multivariate extensions for future work, and it is stated for collapsible effect scales (Section 2). Which baseline characteristics are effect modifiers is taken as given: the framework operates on whatever candidates the sponsor specifies, and their identification, which requires evidence on treatment-by-covariate interaction rather than on prognosis alone (Section 2.4), lies outside it. Because W_1 is non-negative, $\widehat{W}_1$ exhibits positive bias near zero, zero percentile coverage when the true value is zero (S1), and undercoverage near the boundary (S2: 0.703 at $n = 50$) and where the bias is large relative to the sampling standard deviation (S5: 0.862 at $n = 50$), with the intervals erring above the true value (Section 3.2.2);[33] estimates near the null are read against the null floor, and the coverage reported for Study 2 from $n = 100$ per region refers to true distances of about 0.4 to 1.2 SD of the reference distribution, with equal regional sizes; the registered extension (Table 5) covers 0.2 SD at the application's sample sizes, unequal sizes, a heavy-tailed reference and grid-recorded data, but not every combination the application presents. Proposition 1 also assumes a CATE common to all regions; regional differences in extrinsic factors that alter τ itself, such as standard of care or concomitant therapy, lie outside what W_1 measures. Finally, as with any covariate-based approach, unmeasured effect modifiers may contribute to heterogeneity beyond what the distance captures.

The calibration rests on the Lipschitz premise of Proposition 1, and its limitations are those of the premise. First, the constant is a clinical judgment, informed but not determined by prior evidence (Section 2.4), so conclusions drawn from $\Delta_{\max}$, or from the comparison of L^* with L_{UB}, are conditional on it; the same holds for an L_{clinical} taken from an estimated average interaction slope, a lower bound on the Lipschitz constant, for which the safety factor of Section 2.4 is a judgment the framework does not supply. Second, for a CATE that changes steeply over a narrow range of the effect modifier—a threshold or step function—the Lipschitz constant is large and equation (6) is correspondingly loose. This is by design: the bound is a worst case over the CATEs compatible with the stated L_{clinical}, and a loose bound overstates rather than understates the potential difference; the Kolmogorov–Smirnov alternative for that case is stated in Section 2.2. The requirement is also local: since $\bar{\tau}_1 - \bar{\tau}_2 = -\int (F_1 - F_2)\, d\tau$, the bound holds with L_{clinical} replaced by the Lipschitz constant of τ on $\{x : F_1(x) \neq F_2(x)\}$, so saturation of the CATE outside that range does not loosen it, although for distributions with full support, such as normal distributions, that range is the whole line. Finally, the bound is per effect modifier: for a CATE additive across modifiers, $\tau(x) = \sum_k \tau_k(x_k)$, the per-modifier bounds combine as $|\bar{\tau}_1 - \bar{\tau}_2| \leq \sum_k L_k W_1^{(k)}$ and each per-modifier $\Delta_{\max}$ is the contribution attributable to that modifier, whereas under interaction between modifiers they do not, because marginal distances do not control the joint distance. The joint analogue is immediate: if the CATE is Lipschitz with respect to the weighted metric $d(x, y) = \sum_k L_k |x_k - y_k|$, then $|\bar{\tau}_1 - \bar{\tau}_2| \leq W_d(P_1, P_2)$, the Wasserstein distance between the joint distributions under cost d, with or without interaction, and since every coupling of the joint distributions couples each pair of marginals, $W_d \geq \sum_k L_k W_1^{(k)}$: the sum of the per-modifier bounds is a lower bound on the joint one, so that under interaction the per-modifier assessment can understate the heterogeneity. Estimating W_d is a multivariate problem outside the scope of this paper.

Inference tailored to the boundary behavior of $\widehat{W}_1$—resampling fewer than n observations[34] or the numerical delta method for directionally differentiable functionals[35]—could extend the operational range of the intervals at small distances.

The principal contribution of this paper is a distributional similarity measure for effect modifiers whose heterogeneity bound holds for every CATE with a given per-unit clinical slope, is sharp over that class and conservative for any particular CATE, and whose calibration constant is that slope. With bootstrap confidence intervals and clinical calibration via $\Delta_{\max}$ or L^*, W_1 supplies

a reproducible quantitative basis for the judgment of "similar enough" that ICH E17 describes, informing, not prescribing, pooling decisions.

AUTHOR CONTRIBUTIONS

[Author 1] conceived the research question and led the project. [Author 2] developed the mathematical framework and conducted the simulation study. [Author 3] contributed to the application example and manuscript preparation. All authors reviewed and approved the final manuscript.

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FINANCIAL DISCLOSURE

None reported.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

DATA AVAILABILITY STATEMENT

R code for computing the Wasserstein-1 distance and reproducing the simulation studies and the application is available at [repository URL]. The GUSTO-I individual patient data analysed in Section 4 are publicly available as the dataset `gusto` in the R package predtools (version 0.0.3; https://github.com/resplab/predtools); the trial is described in the original publication;[28] the analysis scripts load the data frame with `data(gusto, package = "predtools")` or from the identical CSV copy distributed with the code (N = 40,830 in both). The IST-1 and IST-3 trial databases used in the survey cited in Section 5 are distributed by the University of Edinburgh DataShare (IST-1: https://doi.org/10.7488/ds/104; IST-3: https://doi.org/10.7488/ds/1350, under its data use agreement); the survey protocol, the registered protocols of the Study 2 extension and of the Set 4 control check, and their outputs are included with the R code.

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SUPPORTING INFORMATION

No supporting information accompanies this article; the R code, the registered protocols and all outputs are available as described in the Data availability statement.

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APPENDIX

A PROOFS AND MATHEMATICAL DETAILS

A.1 Non-negativity and Identity of Indiscernibles

Non-negativity is immediate: $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx \geq 0$, since the integrand is non-negative.

For the identity of indiscernibles, we show both directions. If $F_1 = F_2$, then $|F_1(x) - F_2(x)| = 0$ for all x, so $W_1(F_1, F_2) = 0$. Conversely, suppose $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx = 0$. Because the integrand is non-negative, it vanishes for Lebesgue-almost every x, so $F_1 = F_2$ almost everywhere. Right-continuity then upgrades this to equality everywhere: a null set contains no interval, so for any x and every $\delta > 0$ the interval $(x, x + \delta)$ contains points at which $F_1 = F_2$; choosing such $x_j \downarrow x$ and letting $j \to \infty$ gives $F_1(x) = F_2(x)$. Hence $F_1 = F_2$.

Upward bias of $\widehat{W}_1$. Since $E\,\hat{F}_r(x) = F_r(x)$, Jensen's inequality gives $E\,|\hat{F}_1(x) - \hat{F}_2(x)| \geq |F_1(x) - F_2(x)|$ at every x, and integrating (the integrand is non-negative, so no further condition is needed) $E\,\widehat{W}_1 \geq W_1$ for all n_1, n_2.

A.2 Asymptotic Properties

Under standard regularity conditions, the empirical Wasserstein-1 distance $\widehat{W}_1 = W_1(\hat{F}_1, \hat{F}_2)$ is a consistent estimator of $W_1(F_1, F_2)$.[17]

In one dimension, the empirical W_1 distance converges at rate $O(n^{-1/2})$ if and only if $\int \sqrt{F(x)(1-F(x))}\,dx < \infty$.[17] For two samples with $n_1/(n_1+n_2) \to \lambda \in (0,1)$, when $\int \sqrt{F_r(1-F_r)}\,dx < \infty$ ($r = 1, 2$) and $\{x : F_1(x) = F_2(x) \in (0,1)\}$ has Lebesgue measure zero, the directional derivative of W_1 at (F_1, F_2) is linear, which is the condition under which the bootstrap statement below holds;[19] this condition excludes $F_1 = F_2$. At the boundary $F_1 = F_2$ the derivative is non-linear and the limit is not Gaussian; separately, a percentile interval cannot contain zero, which is the degenerate coverage observed under the null (S1; Section 3.2.2). For an effect modifier recorded on a grid, the CDFs are step functions and the condition fails whenever they agree at any grid point inside the support; the finite-space results of Sommerfeld and Munk then apply.[21] Their limit, a maximum over the dual solutions, reduces to a Gaussian when the dual solution is unique, which fails if the two population CDFs coincide on some interval between consecutive grid values. Because the condition concerns the population CDFs, it cannot be checked from the samples; for a distance unresolved against its null floor, such as R2 on systolic blood pressure in Section 4, $F_1 = F_2$ is not excluded and the interval is read at its upper limit only (Section 4.2).

Under the same conditions the percentile bootstrap confidence interval achieves asymptotically correct coverage;[19] where the two CDFs coincide on an interval, including $F_1 = F_2$, the naive bootstrap is not consistent.[20,19]

A.3 Comparison with the Kolmogorov–Smirnov Bound; Attainment on the Outcome Scale

Where both constants are available, neither bound dominates in general. On a support $[a, b]$, $W_1 \le (b-a)\,D_{\mathrm{KS}}$ and $\mathrm{TV}(\tau) \le L_{\mathrm{clinical}}\,(b-a)$, and $L_{\mathrm{clinical}} \cdot W_1 \le \mathrm{TV}(\tau) \cdot D_{\mathrm{KS}}$ holds exactly when $W_1/D_{\mathrm{KS}} \le \mathrm{TV}(\tau)/L_{\mathrm{clinical}}$. For a CATE whose slope has constant magnitude L_{clinical}, such as a linear CATE, $\mathrm{TV}(\tau) = L_{\mathrm{clinical}}(b-a)$ and equation (6) is never the looser of the two; the Kolmogorov–Smirnov bound gains as the CATE's slope falls below its maximum over much of the support, most sharply when the CATE changes steeply over a narrow range—the threshold case in which equation (6) is itself conservative (Section 5). Every valid bound with the constant L_{clinical}, including $L_{\mathrm{clinical}} \cdot W_2$ and, on $[a, b]$, $L_{\mathrm{clinical}}(b-a)D_{\mathrm{KS}}$, is at least $L_{\mathrm{clinical}}W_1$, and the L^* of Section 2.4 is the smallest slope at which a CATE restricted only by its slope produces the margin.

The Kolmogorov–Smirnov bound is likewise sharp within its own class, by a step CATE of height $\mathrm{TV}(\tau)$ at the point where $|F_1 - F_2|$ is largest. A bound on the magnitude of τ itself tightens equation (6) further: on a support $[a, b]$ the attaining τ^* ranges over up to $L_{\mathrm{clinical}}(b-a)$, which on the risk-difference scale is 0.91 for age and 0.56 for systolic blood pressure over the recorded supports of Section 4 (supports that include the records flagged there), exceeding the highest 30-day mortality Lee et al. report for any age group (20.5% above 75 years[27]), so that the attaining CATE would require effects far beyond the baseline risks; for distributions with unbounded support no CATE bounded on the outcome scale attains the bound at all. The 30-day mortality that Lee et al. report for GUSTO-I is 1.1% below 45 years and 20.5% above 75 years,[27] a rise of 19.4 percentage points over at least three decades and hence an average prognostic age gradient of at most about 6.5×10^{-3} per year (Section 2.4).

B R CODE FOR W_1 COMPUTATION

Listing 1 R function for computing the Wasserstein-1 distance with bootstrap confidence intervals.

```
compute_W1 <- function(x1, x2) {
  pooled <- c(x1, x2)
  all_vals <- sort(unique(pooled))
  left <- head(all_vals, -1)
  F1 <- ecdf(x1); F2 <- ecdf(x2)
  w1 <- sum(diff(all_vals) * abs(F1(left) - F2(left)))
  w1
}

W1_bootstrap <- function(x1, x2,
    B = 2000, conf = 0.95) {
  obs <- compute_W1(x1, x2)
  boot_vals <- replicate(B, {
    b1 <- sample(x1, replace = TRUE)
    b2 <- sample(x2, replace = TRUE)
    compute_W1(b1, b2)
  })
  alpha <- 1 - conf
  ci <- quantile(boot_vals,
    c(alpha/2, 1 - alpha/2), na.rm = TRUE)
  list(estimate = obs,
       ci_lower = ci[1], ci_upper = ci[2])
}
```